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Professional Summary

Senior Full-Stack Engineer with 6+ years building production systems across frontend, backend, and AI. At Mastercard, I led architecture for a microfrontend platform serving 10+ product modules, owned backend APIs with Spring Boot and Kafka. Apart from contributor, I am also technical [blogger](#) where I share my learning on various topics such as databases, testing, eBPF, system design .etc.

Completed my Masters at Queen Mary University of London and an [AI fellowship at Nebius Academy](#) in London.

Currently based out of London, UK

Education

Queen Mary University of London

MSc Advanced Computer Science

London, UK

Sep 2025 – Sep 2026

Sardar Vallabhbhai Institute of Technology

BSc Computer Science

Surat, India

Jul 2015 – May 2019

Work Experience

Mastercard

Senior Software Engineer

Mar 2023 – Aug 2025

Vadodara, India

- **Led architecture** for a **microfrontend platform using Webpack Module Federation**, enabling independent deployment of **10+ product modules** with shared component libraries – reduced release coordination from weeks to days.
- **Built full-stack solutions**: frontend in **React and TypeScript** with Redux Toolkit and RTK Query
- Build a framework for customer onboarding with dynamic UIs by building backend APIs in **Spring Boot, JPA, and Hibernate** and leveraging Postgres's JSONB capabilities.
- Build and lead integration of communication of different microservices applying Event sourcing pattern using **Kafka**.
- Shipped **resiliency** by implementing Circuit Breaker and Retry pattern using Resilience4j patterns across 5+ microservices, **improving system availability by 15%**.
- **Drove engineering culture as Accessibility Champion**: lifted **WCAG compliance by 40%**, coached **10+ teams** on best practices, and authored whitepapers adopted by **200+ engineers** with CTO recognition.
- **Owned CI/CD** by writing automation scripts for backend and frontend using Jenkins
- Be part of production triage SRE team to fix outages and production bugs by using observability tools such as Splunk, Dynatrace and Grafana, thereby reducing production MTTR from days to hours.
- **Led cross-functional delivery** as de facto architect, UX designer (Figma prototypes in 2 days), and product liaison – driving alignment across distributed teams and owning end-to-end outcomes.
- Create **Mixpanel** dashboards for product stakeholders to track KPIs such as customer funnels, user engagement .etc.

Mastercard

Software Engineer II

Mar 2022 – Feb 2023

Vadodara, India

- **Built greenfield services** across the stack: **React + TypeScript** frontends with Redux, and **Spring Boot + Oracle** backends with Hibernate; served as core maintainer for performance and code quality.
- **Engineered reliability at scale**: wrote unit tests (Jest, React Testing Library) and e2e tests (Playwright, Cypress) achieving **95% coverage**; reduced CI e2e runtime from **8 hours to 3 hours** via parallelisation and caching.
- **Improved UI performance by 30%** by migrating to an **Akamai CDN** and designing high-fidelity Figma prototypes to align stakeholders in the absence of a UX team.
- Write performance testing scripts in **Gatling and K6**, publishing metrics to **Prometheus** and creating dashboards in **Grafana**

Mastercard

Software Engineer I

Jul 2019 – Feb 2022

Vadodara, India

- **Designed RESTful APIs** to automate BPM workflows using **Camunda**, modelling complex business logic with **BPMN, DMN, and CMMN** standards.
- **Secured payment flows** by developing **Hardware Security Module (HSM) integration** for mobile encryption using **MD5 tokenisation**.

Nebius Academy

AI Engineering Fellow

Mar 2026 – May 2026

London, UK

- **Built and shipped a LangGraph ReAct research agent** that plans its own tool-calling sequence, adding a **dataflow integrity check** to catch hallucinations automatically by verifying claims against tool logs.

- **Designed an MCP (Model Context Protocol) shared tool layer** enabling multiple agent clients to discover and call tools dynamically without code duplication – changing a venue status updates all clients instantly.
- **Shipped a voice pipeline (STT → Agent → TTS)** with graceful degradation when external APIs fail, mirroring production resilience patterns.
- **Implemented LoRA fine-tuning on GPT-2:** froze **124M base parameters**, trained only **~811K adapters (0.65%)**, shifting model behaviour from generic prose to pseudo-Elizabethan diction.
- **Achieved 6.69× LLM inference speedup** via **KV caching**, dropping generation time from **0.943s to 0.141s** for **128 tokens** on H100 infrastructure.

Projects

Repository Summarizer

FastAPI, LLMs, AST, Tree-sitter

- Built a service that extracts code skeletons via AST and Tree-sitter to fit within token budgets, prioritises source files, and assembles context within a 7000-token limit for LLM consumption. Secured admission into the Nebius AI Engineering Fellowship.

Go Microservices

Go, Docker, Kubernetes, gRPC, Envoy

- Progressive microservices project spanning REST APIs, Gorilla Mux with custom validation, Swagger/OpenAPI, Envoy-based service mesh with K8s/WASM filters, and gRPC.

Java GraphQL Service

Spring Boot, GraphQL, JUnit 5

- Production-oriented GraphQL service with query-depth DDOS protection, cursor-based pagination, file upload mutations, DataLoader N+1 resolution, and full JUnit 5 integration tests.

Java CLI (GaalVM)

Java, Picocli, GraalVM

- Java CLI tool using Picocli to query the Stack Overflow API, compiled to a native binary with GraalVM for zero-JVM-startup execution.

Hackathons

Tech: Europe Hackathon— London, UK

Most secure build award

Phantom Fraud

Attio, Tavily, Gemini

- Built a platform bridging deep science and venture capital/biotech by turning raw academic papers into investor-ready pitchdecks.
- Build an AI research agent that expands the query into scientific keywords, searches in publications via Tavily, and returns an instant scorecard which is used to build a brief for investors

Encode Hackathon— London, UK

3rd Place

Phantom Fraud

Solana CLI, OpenRouter, Helius APIs

- Built an **AI-powered financial risk engine** for Solana blockchain to detect wash-trading patterns in real-time.

AI Hackathon — Vadodara, India

3rd Place

AI Code Review Assistant

Gemini Pro Vision, PostgreSQL

- Built a **code-review assistant** powered by fine-tuned Gemini Pro Vision with a PostgreSQL-backed developer UI.

AI Hackathon — Vadodara, India

2nd Place

RAG Documentation Search

Langchain, Chroma Vector DB

- Developed a **RAG-based internal documentation search** using Langchain and Chroma vector database.

Technology Hackathon — Vadodara, India

2nd Place

Next.js Migration

React, Next.js

- Migrated a React app to **Next.js**, improving page performance by 35%.

Technical Skills

Frontend: React, TypeScript, Next.js, Redux Toolkit, RTK Query, Webpack, Module Federation, Tailwind CSS, Figma

Backend: Node.js, Python, Spring Boot, FastAPI, Go, REST APIs, GraphQL, Microservices, Event Sourcing, Kafka

AI/LLM: LangGraph, ReAct, MCP, RAG, Vector DBs, PyTorch, Transformers, LoRA, Multi-Agent Systems, Fine-tuning

Data & Storage: PostgreSQL, Oracle, Redis, MongoDB, Chroma, Parquet

Cloud & DevOps: AWS, GCP, Docker, Kubernetes, Terraform, Jenkins, GitHub Actions

Observability: Grafana, Prometheus, OpenTelemetry, Splunk, Dynatrace

Testing: Jest, React Testing Library, Cypress, Playwright, JUnit, Mockito, Testcontainers

Practices: Agile/Scrum, TDD, Pair Programming, Code Review, OWASP Top 10, Design Patterns

Writing, Talks & Recognition

40+ technical articles on system design and architecture (aymanace2049.hashnode.dev). Featured in the *eBPF newsletter*.

Company-wide whitepapers on Accessibility and Microfrontend architecture; earned CTO recognition, adopted as internal reference material by **200+ engineers**.

Kafka Seminar (Oct 2018): Delivered a technical seminar on Event-Driven Architecture using Apache Kafka, covering Kafka Architecture, Kafka Streams, Kafka Connect, and AMQP messaging patterns. [Report](#)